

The Perceived Consequences of Gold Mining in Postwar El Salvador: A Qualitative Study

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Mining activities raise significant public health concerns in the Americas and globally.^{1–3} Many countries in Central America have experienced increased mining activity by national and foreign corporations in the wake of trade agreement liberalization and erosion of environmental protections.^{4,5} This increase in mining activities has led to **tension between mining corporations and local populations concerned with the health effects of mining**.⁶ Nongovernmental organizations have begun to document such conflicts, which currently affect 19 countries from South to North America, with 191 ongoing mining conflicts.⁷ Case reports from other countries undergoing metal mining (e.g., gold, silver, and nickel) reveal significant health effects from mining. These range from hearing loss and mercury poisoning⁸ to elevated arsenic levels in the blood of local inhabitants.⁹

Although these toxicological studies have delineated direct threats to health from metal mining, little if any research in Latin America has explored other detrimental effects on public health. In addition, the effects of mining in the historical context of civil war and foreign intervention in Latin America have not yet been explored. We therefore undertook a qualitative inquiry of a representative Latin American country, to explore attitudes about the public health ramifications of metal mining in a postwar context.

El Salvador is unique in Latin America for historical and demographic reasons. The population of El Salvador experienced a civil war from 1980 to 1992. Up to 75 000 Salvadorans died (out of a population of 5.2 million), and a third of the population were internally displaced or fled the country.¹⁰ The civil war in El Salvador received extensive funding from foreign countries, primarily the United States, which contributed to the ongoing conflict and its associated deaths and disappearances among the local population.^{11,12} The mental health impact of the civil war is only now being understood.¹³

Objectives. We investigated themes related to the health and environmental impacts of gold mining in El Salvador.

Methods. Over a 1-month period in 2013, we conducted focus groups (n = 32 participants in total) and individual semistructured interviews (n = 11) with community leaders until we achieved thematic saturation. Data collection took place in 4 departments throughout the country. We used a combination of criterion-purposive and snowballing sampling techniques to identify participants.

Results. Multiple themes emerged: (1) the fallacy of economic development; (2) critique of mining activities; (3) the creation of mining-related violence, with parallels to El Salvador's civil war; and (4) solutions and alternatives to mining activity. Solutions involved the creation of cooperative microenterprises for sustainable economic growth, political empowerment within communities, and development of local participatory democracies.

Conclusions. Gold mining in El Salvador is perceived as a significant environmental and public health threat. Local solutions may be applicable broadly. (*Am J Public Health.* 2015;105:2382–2387. doi:10.2105/AJPH.2015.302832)

El Salvador has among the highest population densities in the Western Hemisphere. Thus, the traumatic effects of disasters such as war, earthquakes, or environmental degradation have significant impact on large sectors of the population. Past exposure to such disasters—natural and manmade—have perhaps sensitized the population to the potential for extensive harm from future disasters.¹⁴ This has yet to be formally explored in a Salvadoran context.

Since the end of the civil war in 1992, gold mining activities have increased in El Salvador, with public health concerns increasing in parallel. This may explain, in part, why local opinion polls reveal concern about mining activities. In 2007, the Instituto Universitario de Opinión Pública (University Institute of Public Opinion) of the Central American University conducted a poll among citizens living in metal-rich areas, typically in the north of the country. It found that **70% of Salvadorans opposed the initiation of a mining project in their community and felt this would have a detrimental impact on the lives of their children and grandchildren**.¹⁵ Because limited

data are available in the medical literature about the population health effects of mining specifically in El Salvador, it is unclear exactly how mining activity would be detrimental.

Although the political, social, and historical context of El Salvador is unique in the region, many parallels exist with neighboring countries. **We explored Salvadoran community leaders' attitudes toward gold mining in a post-war context.** We also wished to understand **solutions and alternatives to mining that might be proposed by community leaders.** Their attitudes, although specific to El Salvador, might be relevant to other countries in Latin America with a similar history and increases in mining-related conflict. The lessons learned from the experiences shared in El Salvador may therefore be relevant to countries beyond Central America.¹⁶

METHODS

We used a combination of **criterion-purposive and snowballing sampling techniques** to identify potential participants. We

approached leaders of social or economic governmental and nongovernmental organizations who were visible in the local, national, or international media discussing mining issues. We elicited multiple viewpoints from these various stakeholders (from agronomists to lawyers) to gain insight into mining practices in El Salvador.

Interviewing

Interviews took place in several geographic areas of El Salvador over the month of May 2013. These included 2 territorial units (departments) in the north, which were directly affected by mining practices (Chalatenango and Cabañas), in addition to San Vicente in the center. Additional focus groups and individual interviews took place in the department and capital of San Salvador, where the main headquarters of some organizations were located. We conducted all interviews in Spanish.

We asked participants to describe their ideas about mining as a tool for economic development, perceived negative and positive effects, mining in the context of postwar El Salvador, and potential solutions and alternatives to mining. "Tell us about the role of mining in El Salvador with respect to . . ." was a common opening stem, with exploration of the overall role in, among other areas, development, public health, and history. If an interviewee provided a perspective that was either overtly negative or positive toward the role of mining in El Salvador, we would recast our semistructured questions to explore, in an open-ended fashion, any other opinions.

Violence has occurred against some members of environmental and social groups in El Salvador who may be considered to be opposed to the corporate interests of foreign mining investors. This violence has included death threats, torture, and assassinations.¹⁷ Most cases have remained unsolved, or minor actors have been accused or convicted of such activities. For these reasons, we kept both the participants and the organizations they represented anonymous.

Data Collection and Analysis

Our data collection and management followed the practices of Krueger.¹⁸ We taped and later transcribed all focus groups and individual interviews. The main interviewers (T. L. Z

and P. C.) were trained in qualitative methods and used a semistructured interview guide. We used a combination of grounded theory methodology and phenomenological inquiry for our qualitative analysis. We identified codes and themes from grounded observation and data, while focusing on participants' perceptions, perspectives, and understandings of phenomena.^{19,20}

The main interviewers recorded all interviews with permission and transcribed them with manual coding of emergent themes. We used a constant comparison analysis approach to ensure interpretive rigor.²¹ Through this comparison and discussion, we developed a coding frame, which we then applied to all data. We organized each theme or topic into larger categories as the research proceeded. Finally, we grouped the codes together into themes with various subthemes, which constituted the foundation of synthesizing and conceptualizing the relationships among the data.

Both focus groups and individual interviews continued until thematic saturation occurred and no new themes emerged from the interviews. Although interviewees offered, at times, somewhat divergent ideas, we had no deviant cases or enough contradictory evidence to juxtapose these with the main themes.

In El Salvador, the research was not aligned formally with an academic institution, but rather with the largest rural and national nongovernmental organization, which represented more than 300 communities throughout El Salvador, the Association for the Development of El Salvador (CRIPDES).²² Its mandate is to improve the development, employment, and living conditions of rural Salvadorans.

RESULTS

We conducted 6 focus groups representing a total of 32 community leaders over a 1-month period in 2013. We also conducted 11 individual interviews during the same period. Participants were full-time community leaders, agronomists, *campesinos* (farmers), health promoters, and lawyers. All were Salvadoran. The majority of the participants were male (91%), and many had witnessed and survived the civil war. Four major themes emerged regarding the impact of mining on the population of El Salvador (see the box on

the next page). Mining held the promise of economic activity and development through job creation and foreign development opportunities. According to our respondents, these benefits did not materialize (theme 1). Those who were critical of mining activities cited environmental and health concerns (theme 2). Tension within communities arose between supporters and critics of mining, and some critics experienced violence (theme 3). Participants expressed agency and empowerment through several solution-focused strategies to improve mining-independent economic development in El Salvador (theme 4).

The Fallacy of Economic Development

A common theme emerged regarding the question of mining as a tool for development. Virtually all participants considered the promise of national economic growth through mining to be false. They did not perceive mining by foreign companies in El Salvador as a sustainable and beneficial economic activity and thus rejected mining as a path to development. Participants viewed mining as unsustainable in the long term and an overall economic detriment to the country and to Salvadorans.

Fallacy of local job creation and economic growth. Our respondents did not believe that mining would encourage local economic growth (through direct and indirect job creation). Participants said that the permanent, well-paying positions would be reserved for foreign technicians, because El Salvador has a paucity of local experts on the technical, logistical, and engineering aspects of mining. Respondents considered the manual labor positions that would be available to be precarious employment: dangerous, temporary, and low paying (US \$7/day).

Another concern was that public land development to assist in mining exploration (e.g., the creation of public roads leading to mining projects) would be a further economic drain on the public sector.

Fallacy of national development. Many participants said that international and national neoliberal trade laws favored foreign investment over the well-being of the Salvadoran population. Most participants reiterated that the 2% or lower corporate tax rate would not benefit the nation's economy. All participants said that the economic benefits, in the context

Representative Quotes From Participants in Qualitative Study on Public Health Effects of Gold Mining: El Salvador, 2013

Theme 1: The Fallacy of Economic Development

"Mining is just a temporary economic activity. We are not skilled for the engineering or technical components. It only lasts as long as there are metals to exploit. Once that is gone, well the story ends there. Maybe we will work 1 year, or 2 doing menial labor, then we are out. So we are not talking about long-term work. That is not, by any means, sustainable employment."

"International mining corporations are experts at knowing how to evade paying taxes. It's not even illegal. The laws of El Salvador established after the war made it favorable for foreign corporations to invest here. They can even sue our government, and they are, if they feel they have been prevented from capturing all the profit they are able to collect."

"Development? It is also already so easy for corporations to avoid paying the ridiculous 2% corporate tax rate. There is barely any contribution made to our country as they remove the minerals from our lands. Meanwhile all we are left with is the environmental mess to clean up, for generations, once they are gone. How is that 'development'?"

"The concept of development for me is not a product that can be exported, you can't measure it like that. The concept of development is in the community, in the way people think, see, cultivate and support ideas. That is the way we develop. A municipality like ours . . . we are based in agriculture, in harvesting to develop. If large industries are brought in, with large machinery, we can't operate those, it will be foreigners that will do it. They only thing we can do is suffer the damages. If it's considered "development," then development for whom? The corporations or the community?"

Theme 2: Critique of Mining Activities

"Mining, by nature, is neither sustainable nor benefits nature."

"Where there is gold, there is arsenic."

"We are a high-density population that is already water insecure. Mining will lead to the contamination of our aquifer with arsenic, lead, and sulfur. The process of mining requires millions of liters of water to extract just a small quantity of gold. We are left with the contaminated water afterwards and that will affect the health of Salvadorans for generations."

"I've seen kids with skin cancer. They start with a small sore then it spreads throughout their whole body. There was a 10-year-old girl who died, she couldn't move anything, just her eyes, and then she just stopped breathing. Even the baby pigs. We saw a pig with 7 piglets. Only 3 of them could walk properly."

Theme 3: Creation of Mining-Related Violence: Parallels to the Civil War

"When we found Marcelo's body . . . that was hard. That was very hard. His fingernails had been pulled out . . . and his hand had been shoved into his mouth. His genitalia had signs of torture. Forensically speaking, this was the same pattern of torture used by the death squads during the civil war. The sole purpose was to intimidate the community" [referring to the 2009 death of Marcelo Rivera, a well-known community leader active in raising awareness about the negative health and environmental effects of mining].

"The same mechanisms of repression and instilling terror in people to stop resisting mining activity—those same mechanisms were the very ones used during the civil war."

Theme 4: Solutions and Alternatives to Mining Activity

"We have identified the need to have economic control over our lands—together. Many communities have started working in microenterprises and cooperatives, together. The workers have the ultimate say on how to run the industry, whether it's for cashew production or a local fishery. This economic power grows and we bring this model of participatory democracy into the educational system of the community, the health care units, and so on. We all participate, we support each other, from the ground up. The mining corporations feel threatened because it's a different model than what they use."

"It's more than just having employment alternatives to mining, it's about having a future for Salvadorans created by Salvadorans, where all perspectives are respected. We don't want a present that is harmful and dictated by outside interests for their benefit only."

Note. These interviews were conducted in 4 departments throughout El Salvador: Chalatenango, Cabañas, San Vicente, and San Salvador. T. L. Z. and P. C. translated the interviews from Spanish to English.

of current trade laws, disproportionately favored the investing (typically high income) country and not the host (low income) country.

Participants highlighted the high cost of postmining environmental cleanup that would be required over years or decades, long after any short-term economic gain could be realized. The most common theme was a need for the clarification of the idea of development. To our respondents, this entailed a future in which Salvadorans would be able to work with dignity, earn a living wage, and do so in an environmentally sustainable (both locally and globally) manner, while directly participating in democratic decision-making regarding the future of their country. Mining was not viewed as a part of this vision.

Critique of Mining Activities

Environmental effects. The concept of *estrés hídrico* (water stress) emerged as a predominant theme. Only 3% of the superficial water in El Salvador was of sufficiently safe quality to drink, and most participants viewed mining to be a major threat to the aquifer system of the country. Participants expressed concern about both the vast quantity of water used during the mining process itself and the concomitant risk to waterbeds posed by tailings or harmful chemicals.

Respondents cited previous experiences with mining and water contamination. The most commonly mentioned chemical released by the mining process was arsenic, which raised significant concern for harm to the Salvadoran

population. Participants also expressed frustration at the lack of government testing of water and land samples in areas currently undergoing mining exploration.

Health effects. Health effects of mining on both humans and animals were a major concern to our respondents, particularly the possibility of reduced access to clean, drinkable water. Themes related to human cases of dermatological and neurological conditions, such as Guillain-Barré syndrome. Participants also cited cases of exposed cows and newborn calves that were unable to walk and of hematologic cancers that disproportionately affected rural area populations.

Another theme was participants' concerns about the high population density of El

Salvador. They stated that if mining contaminated the land, this would inevitably and negatively affect the health status of Salvadorans. Many contrasted this scenario to mining in a country such as Canada, where the population density is far less, so that our respondents considered the health repercussions and environmental effects to be less significant.

Mining-Related Violence

Participants viewed criticizing the health, environmental, and economic impact of mining in El Salvador as dangerous, placing community leaders at risk for violence. They described how different forms of violence have affected many Salvadorans in ways that paralleled the violence of the civil war. Respondents believed that this violence emanated from forces and actors that supported mining activity in the country. The purpose of this violence, according to our respondents, was to terrorize community leaders and, by extension, communities, to intimidate them into allowing mining exploration and exploitation. This violence affected individuals, communities, and the country as a whole, including effects on the stability of the government. Participants reiterated that in parallel to the civil war, mining activity received support from foreign forces that threatened the autonomy of the nation, a form of foreign intervention. Resistance to such violence was created through strengthening community ties and activities.

All participants highlighted the threat of violence arising from their environmental- and mining-related activities. They described direct threats of violence, in particular, torture and assassinations perpetrated in the north of the country against men and women involved in perceived antimining activism in 2009 to 2011. Many participants mentioned the death by torture of antimining activist Marcelo Rivera in 2009. They described death threats as commonplace for various members of the community perceived to be sympathetic to environmental concerns and viewed as unsupportive of the mining industry.

Displacement of populations. A parallel theme was the internal displacement of populations attributable to mining activities. Populations, typically rural, were encouraged to leave or to sell lands thought to be mineral rich.

Participants said that mining interests usually accomplished this through a combination of threats and bribes, with threats more common because of the low proportion of *campesinos* who owned the land they worked. Many participants noted similarities with forced civilian displacement during the war. Respondents cited this further displacement as a profound source of collective emotional and psychological stress for the population.

Social divisions. Mining companies, according to many participants, also encouraged social divisions. This included direct financial support for selected family members to ensure loyalty to corporations engaged in mining. This tactic was particularly notable in communities or families that had some members known to openly oppose mining activities.

Participants also described social divisions on different political levels. At the municipal level, mining interests encouraged tension and violence by financially backing low-level politicians or community leaders perceived to favor mining, such as clergy. Many participants said that community cohesion suffered in towns and rural areas where mining activities had been the most active. They found these types of social divisions in Salvadoran society to be similar to the divisions that deepened during the civil war.

Structural violence. Several participants considered that the tension between polarized groups in El Salvador—groups either supportive or critical of mining practices—was nothing more than a continuation of the civil war. Many noted that the same structures of power and exclusion that perpetuated war in the 1980s remained in place to facilitate what some called the current war of exclusion. Several respondents described structural violence as intrinsic to the well-entrenched laws and neoliberal economic model of El Salvador, which favored the rich over the poor. During the civil war, a domestic economic elite formed the government and enjoyed financial backing by a foreign entity (the US government). During the current war of exclusion, an economic and corporate elite (typically foreign mining corporations) was disproportionately favored. Participants blamed Salvadoran neoliberal laws that, by design, encouraged corporate endeavors and

wealth generation over democracy, population health, and well-being.

Some participants believed that the power of foreign elites, much like during the civil war, was so great that Latin American governments that challenged mining corporations risked military coups and dissolution of democracy. The case of Honduras, where President Manuel Zelaya was ousted by a military coup d'état, was frequently provided as an example.

Solutions

Solutions were not only proposed but also implemented by individuals, communities, and local populations that resisted mining activities. Examples given by our respondents ranged from local economic initiatives to a broader strengthening of popular power. This extended even beyond traditional political party lines, to support national autonomy in economic decision-making. Participants considered many of these solutions to be not reactive in nature but rather proactive: designed as a new way that Salvadorans could work cooperatively. Participants also reported another feature of these activities: decision-making in local industries that were organized as cooperatives was Salvadoran and collective, rather than conforming to a foreign and top-down corporate structure.

Economic empowerment. Economic empowerment was important for communities and leaders active in resistance to mining. Because mining was the main, immediate source of income for some, participants emphasized the importance of economic alternatives. Examples provided were the creation of cooperatives in the form of community, women-organized *cafeterias*; small-scale local textile factories that produced school uniforms; and larger-scale enterprises that engaged in fishing and preparing cashews for exportation.

Participants said that it was important to have financial independence, but with income generation from communal activities. This principle of community through economic activity provided alternatives to mining for community members in need of work. Such alternatives generated economic independence and a new form of economic interaction.

Control of land and communities. Economic autonomy fostered economic empowerment and decision-making. Many participants gave examples of a form of resistance to mining that

involved purchasing exploitable mineral-laden land and converting it to religious sites, rendering it unavailable for mining for deep, traditional reasons.

Increased economic activity and increased autonomous control of lands created a positive feedback loop for community strength and empowerment that went beyond resistance to the creation of democratic and collective, community-based solutions to other local challenges. For example, participants explained how local schools were created that, among other things, generated knowledge and incorporated the effects of mining into the local curriculum, together with methods of resistance. Foreign-based mining practices in certain communities have, in fact, created a stronger base of participatory democracy by arousing the local population to action. Participants were quick to point out that this occurred predominantly in communities that had a history and tradition of autonomy and resistance that persisted throughout the war and beyond.

DISCUSSION

Ours was the first study to explore the perceived effects of gold mining in Latin America, specifically within a postwar context, from the perspective of individuals most engaged in the issue. Our respondents considered the impact of gold mining to be much broader than direct health or environmental effects, believing that it imposed a foreign-backed structure of violence. Participatory and democratic activities on local levels were proposed as ways to abrogate this violence. Respondents pointed out many similarities between mining-related violence and the violence and trauma experienced by Salvadorans during their civil war. This may be similar to experiences in other countries in Latin America (and elsewhere), in light of the recent surge in metal mining and the history of foreign-backed violence during times of war and intervention (e.g., in Chile, Colombia, Guatemala, and Peru).²³

Participants described not only violence that was very reminiscent of events during the war (direct, individual violence and torture), but also a deeper, structural form of violence in which the national, democratically elected government experienced pressure from

external forces (mining companies and the foreign governments that benefited from their profits, e.g., Canada). Frequently mentioned was a lawsuit filed by Pacific Rim of Canada against the Salvadoran government for more than \$300 million because the company was denied a license for gold exploration.²⁴ Of utmost concern to our respondents were cases of direct human rights violations involving torture and assassinations and subsequent dissatisfaction with police investigations into these crimes.

We identified solutions that were used to resist mining activities and strengthen community empowerment. This largely took the form of creative microenterprises and collective, community-based cooperatives. This economic activity led to increased financial independence and an alternative to mining as an employment option. Once communities were strengthened economically, control over the land and decision-making also increased, leading to an increase in community-based participatory democracy, both locally and regionally, which in turn supported the national government in making autonomous decisions (e.g., placing a moratorium on mining until the public health ramifications could be explored). This increase in participatory democracy on local levels, in autonomous zones, was reminiscent of events during the civil war.

Limitations

We did not independently verify our informants' multiple claims of toxic exposure, birth defects, human rights violations, and other harms. These claims may or may not be true. Nonetheless, our purpose was not to verify the factual nature of claims or test their validity. The purpose of qualitative research is to explore the perspectives of participants, regardless of whether their claims are true or false. Subsequent studies or investigations could substantiate these claims.

Information gathered from focus groups represented the attitudes and perceptions of a narrow sample of the population. However, these perceptions are highly valuable because they will inform communication strategies dealing with mining development. We selected our participants because of their work in and knowledge of the environmental and health issues of mining. This may not have been

representative of the perspectives of the Salvadoran population as a whole. Without further inquiry, generalizations regarding larger populations must be avoided.

Conclusions

Our study provides new insight into Salvadorans' perceptions of the public health effects of gold mining, in a postwar context. Respondents expressed skepticism about the economic development claims made by proponents of mining and described the violence perpetrated against those who opposed mining.

Participatory democracy and economic empowerment are alternatives required to resist the effects of mining activities (and associated violence). Our data may be reflective of other Latin American countries that have similar experiences with mining, also in a context of war and conflict. ■

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Contributors

T. L. Zakrison and P. Cabezas conducted the interviews and developed the article. E. Valle transcribed the interviews. J. Kornfeld and C. Muntaner developed the conceptual framework and planned the project. S. Soklaridis assisted with the data analysis. All authors revised the article.

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Human Participant Protection

The University of Miami human subjects research office approved the study protocol.

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